

Adam Valade

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EDUCATION

University of Michigan — B.S. Computer Science

Expected December 2026

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Web Systems, Database Systems

TECHNICAL SKILLS

Languages: C++, Python, JavaScript/TypeScript, SQL

Systems: REST APIs, distributed systems, relational data modeling

Frameworks: React, Flask, Express

Databases: PostgreSQL, MongoDB, Redis

Tools: Git, Docker, AWS, Postman

Concepts: distributed systems, concurrency, fault tolerance

SELECTED ENGINEERING PROJECTS

ForgeQueue — Fault-Tolerant Distributed Job Queue (Python / Redis / Docker / REST)

Live: <https://forgequeue-api.onrender.com/demo/> | **Github:** github.com/Adamvalade/forgequeue

- Built a Redis-backed distributed job queue with at-least-once delivery semantics, worker leasing, visibility timeouts, and crash recovery.
- Implemented retry orchestration with exponential backoff, delayed scheduling, dead-letter queue handling, and requeue support.
- Designed idempotency, deduplication, and per-job locking to reduce duplicate processing under concurrent worker execution.
- Exposed REST APIs for job submission, status tracking, and operational metrics, and containerized the system with Docker for reproducible deployment.

Cadence — Full-Stack Music Review & Social App (Next.js, FastAPI, PostgreSQL, Redis)

Live: cadencemusik.com | **Github:** github.com/Adamvalade/Cadence

- Built and deployed a full-stack music review platform with auth, ratings, reviews, lists, and profiles
- Designed REST APIs in FastAPI with async SQLAlchemy, JWT-based authentication, and rate limiting for core application flows.
- Developed a responsive Next.js frontend with dynamic pages for album discovery, user libraries, and social interactions.
- Architected and deployed a multi-service system with PostgreSQL, Redis, API, and frontend containers using Docker Compose.

Chess Engine — C++ / Python / JavaScript / REST API

Live: megatronchess.com | **Github:** github.com/Adamvalade/megatron-chess

- Implemented a high-performance bitboard-based move generator in C++ supporting legality checks, castling, en passant, check, and checkmate.
- Integrated the C++ engine core with Python via ctypes to combine low-level move generation with higher-level search and evaluation logic.
- Built a browser-based chess interface with real-time move validation, animation, and game-state synchronization through a REST API.

EXPERIENCE

Data Operations Intern

Alta Vista Technology

2023–Present

- Automated CRM workflows to enforce data consistency and reduce manual intervention.
- Built validation and reporting logic to detect and correct duplicate or malformed records.
- Integrated CRM data with downstream financial systems via API-based synchronization.
- Diagnosed data pipeline failures and documented resolution patterns.

Developer

Michigan Build & Launch

2024–Present

- Built modular React components and dynamic page-builder tools for a Weebly-style site creator.
- Implemented backend logic for user authentication, data persistence, and RESTful API routes.
- Contributing to architecture planning and sprint execution in a 6-person engineering team.
- Translated product requirements into implementable technical designs during sprint planning

